# Skills Summary

## Making a Piecewise Model Continuous

*Use this reference while completing your worksheet or when studying.*

---

**Continuity at a junction $x = c$ means three things agree:** the left-hand limit, the right-hand limit, and the value $f(c)$. Every problem on this topic is the same move — write the two one-sided limits at $c$, set them equal, and solve for the unknown constant.

## Skill 1 — Matching the Two Sides at a Junction

**Method:** substitute the junction input into *each* piece separately. The piece for $x < c$ gives the left-hand limit, the piece for $x \geq c$ gives the right-hand limit. Set the two expressions equal and solve. The unknown appears in exactly one of them, so the equation is linear.

**Example:** $f(x) = 2x + a$ for $x < 4$ and $f(x) = 3x - 5$ for $x \geq 4$. Find $a$.
**Step 1.** Continuity at 4 means the two pieces must hit the same height there, so evaluate each piece at 4 and set them equal.
**Step 2.** Left: $2(4) + a = 8 + a$. Right: $3(4) - 5 = 7$. So $8 + a = 7$, giving $a = -1$.
$\Rightarrow$ $\boldsymbol{a = -1}$

**Try it:** $f(x) = 4x + a$ for $x < 3$ and $f(x) = 2x + 7$ for $x \geq 3$. Find $a$.

check: $\boldsymbol{a = 1}$

## Skill 2 — Filling a Hole With the Limit Value

**Method:** when a piece is a quotient that gives $\frac{0}{0}$ at the junction, the discontinuity is a hole. The only value that removes it is the limit, so compute the limit — factor and cancel, or multiply by the conjugate — and set the constant equal to it.

**Example:** $p(x) = \dfrac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $p(2) = c$. Find $c$.
**Step 1.** The formula is undefined at 2, so $c$ has to be whatever the limit approaches — compute the limit first, then read off $c$.
**Step 2.** $\dfrac{x^2 - 4}{x - 2} = \dfrac{(x-2)(x+2)}{x - 2} = x + 2$ for $x \neq 2$, and at $x = 2$ that is 4.
$\Rightarrow$ $\boldsymbol{c = 4}$

**Try it:** $q(x) = \dfrac{x^2 - 36}{x - 6}$ for $x \neq 6$ and $q(6) = c$. Find $c$.

check: $\boldsymbol{c = 12}$

## Skill 3 — Two Junctions, Two Unknowns

**Method:** a three-piece model has two junctions, so it gives two equations in the two unknowns. Write both, then eliminate: solve one for $b$, substitute into the other, and back-substitute.

**Example:** $f(x) = x^2$ for $x < 1$, $ax + b$ for $1 \leq x < 3$, $2x + 5$ for $x \geq 3$. Find $a$ and $b$.
**Step 1.** Each junction gives one equation, and two unknowns need two equations — so write both junctions before solving either.
**Step 2.** At 1: $a + b = 1$, so $b = 1 - a$. At 3: $3a + b = 11$, so $3a + (1 - a) = 11$, giving $2a = 10$ and then $b = 1 - 5$.
$\Rightarrow$ $\boldsymbol{a = 5}$ and $\boldsymbol{b = -4}$

**Try it:** $f(x) = 3x$ for $x < 2$, $ax + b$ for $2 \leq x < 4$, $x^2 + 2$ for $x \geq 4$. Find $a$ and $b$.

check: $\boldsymbol{a = 6}$, $\boldsymbol{b = -6}$